

Formula Sheet

P = principal/present value r = annual interest rate m = compoundings per year $n = tm$
 A = future/maturity value t = number of years $i = r/m$ = rate per period r_E = effective rate

Simple Interest	Compound Interest	Continuous Compounding
$A = P(1 + rt)$	$A = P(1 + i)^n$	$A = Pe^{rt}$
$P = \frac{A}{1 + rt}$	$P = A(1 + i)^{-n}$	$P = Ae^{-rt}$
	$r_E = \left(1 + \frac{r}{m}\right)^m - 1$	$r_E = e^r - 1$

R = periodic payment P = present value of annuity n = number of periods
 S = future value i = rate per period

Future Value of an Annuity	Sinking Fund Payment
$S = R \left[\frac{(1 + i)^n - 1}{i} \right]$	$R = \frac{S \cdot i}{(1 + i)^n - 1}$

Present Value of an Annuity	Amortization Payments
$P = R \left[\frac{1 - (1 + i)^{-n}}{i} \right]$	$R = \frac{P}{[1 - (1 + i)^{-n}]}$

Ti-83/84

N : number of payments
I% : interest rate
PV : present value
PMT : payment size
FV : future value
P/Y : payments per year
C/Y : compoundings per year

You must show work to receive credit.

Find the simple interest AND the future value of the loan.

1. \$1000 at 4.25% for 6 months.

Find the compound amount for the given deposit and the amount of interest earned.

2. \$2500 at 6.01% compounded monthly for 38 months.

Find the amount that should be invested now to accumulate the following amount, if the money is compounded as indicated.

3. \$6200 at 9.5% compounded quarterly for 29 months.

Solve

4. A company will need \$100,000 in 9 years for a new building. To meet this goal the CFO will invest money into an account paying 9% annual interest rate compounded quarterly. How much money needs to be invested into the account?
5. You place \$1500 into an account which pays 5% compounded daily. How much money will be in the account after 800 days? Assume that there are 360 days in a year and 30 days in a month.

Find the future value of each ordinary annuity, if payments are made and interest is compounded as given. Then determine how much of this value is from interest.

6. $R = \$3,250$ (payments); 11% interest compounded semi-annually for 7 years.

Determine the interest rate needed to accumulate the following amounts in a sinking fund, with payments as given.

7. Accumulate \$56,000, monthly payments of \$300 over 12 years.

Find the periodic payment that will amount to each given sum under the given conditions.

8. $S = \$100,000$; interest is 5% compounded annually; payments are made at the end of each year for 12 years.

Find the future value of each ordinary annuity, if payments are made and interest is compounded as given. Then determine how much of this value is from interest.

9. Payments of \$800 into an account paying 6.51% interest compounded quarterly for 12 years.

Determine the interest rate needed to accumulate the following amounts in a sinking fund, with payments as given.

10. Accumulate \$56,000, monthly payments of \$300 over 12 years.

Find the present value of each ordinary annuity.

11. Payments of \$100 each month for 10 years at 8.5%.

Find the monthly house payments necessary to amortize each loan. Then calculate the total payments and the total amount of interest paid.

12. \$175,000 at 6.24% for 30 years.